

	(since emf remains the same). This will further increase the temperature of the thermistor if it is not kept cooled.
4(a)	One tesla is the magnetic flux density when a conductor of length 1 m carrying a current of 1 A, placed perpendicular to the magnetic field, experiences a force of 1 N.
4(b)(i)	As the particle enters the magnetic field at right angles, it will experience a magnetic force that is at right angles to both the magnetic field and its velocity. This magnetic force provide the centripetal force for the motion of along the arc of a circular path.
(ii)	$F_B = F_C$

$$qvB = \frac{mv^2}{r}$$

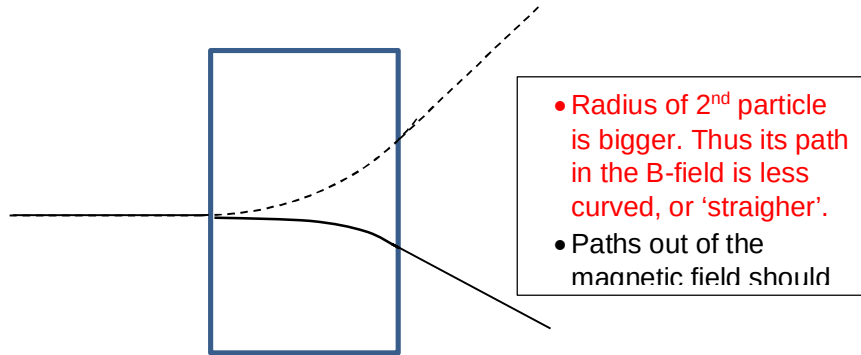
$$qB = \frac{mv}{r}$$

$$p = rqB$$

$$= (6.2 \times 10^{-2}) (1.6 \times 10^{-19}) (0.24)$$

$$= 2.4 \times 10^{-21} \text{ N s}$$

4(c)



5(a)

A photon is a quantum of electromagnetic radiation with energy given by hf ,